

# Skills Summary

## Radius or Diameter? Length or Covering?

Use this reference while completing your worksheet or when studying.

### Skill 1 — Halve a diameter before you square it

**Radius**  $r$  reaches from the centre to the edge. **Diameter**  $d$  crosses the whole circle through the centre, so  $d = 2r$ .

$$C = 2\pi r = \pi d \quad (\text{either number works}) \quad A = \pi r^2 \quad (\text{radius only})$$

Classify the given number first: “across”, “through the centre”, “wide” → diameter. “From the centre” → radius.

**Example:** A circle is 10 cm across at its widest point. Find its area.

**Step 1.** “Across at its widest point” is a diameter, and  $A = \pi r^2$  accepts a radius only, so the halving has to happen before the squaring.

**Step 2.**  $r = 10 \div 2$ , so  $A = \pi(10 \div 2)^2 = \pi \times 25 = 78.5398\dots$  The area is **78.54** cm<sup>2</sup>.

**Try it:** A circle is 6 cm across at its widest point. Find its area, rounded to the nearest hundredth.

check: 28.27 cm<sup>2</sup>

### Skill 2 — Around the circle, or across its face?

**Circumference** is the trip around the rim: a *length*, measured in cm or m. **Area** is the face the circle covers: a *covering*, measured in cm<sup>2</sup> or m<sup>2</sup>.

Deciding words: *ribbon, fence, trim, rolls once around* →  $C = 2\pi r$ . *Paint, paper, covers, surface* →  $A = \pi r^2$ . The unit of your answer must match the word you picked.

**Example:** A circular badge has radius 4 cm. How much ribbon runs around its edge?

**Step 1.** “Around its edge” names a length, so this is circumference — the squared formula would answer a question nobody asked.

**Step 2.**  $C = 2\pi r = 2\pi(4) = 25.1327\dots$  The ribbon needed is **25.13** cm.

**Try it:** A circular badge has radius 8 cm. How much paper does it take to cover the badge’s face?

check: 201.06 cm<sup>2</sup>

### Skill 3 — Same rectangle, two different questions

For a rectangle with sides  $a$  and  $b$ :

**Perimeter**  $P = 2a + 2b$  — add your way around the outside (a length).

**Area**  $A = a \times b$  — multiply to cover the inside (a covering).

A composite shape has no single formula: cut it into rectangles, find each area, and add. Never add side lengths when the question asks what the shape covers.

**Example:** A deck is 7 m long and 3 m wide. How much railing runs around it?

**Step 1.** Railing follows the outside edge, so the question wants perimeter — the adding formula, not the multiplying one.

**Step 2.**  $P = 2(7) + 2(3) = 14 + 6$  The railing needed is **20** m.

**Try it:** The same deck is 7 m long and 3 m wide. How many square metres of decking board cover it?

check: **21 m<sup>2</sup>**

**(!) Watch out:** Forgetting to halve a diameter multiplies an area by four, not by two, because the radius is squared. And an answer whose unit disagrees with the formula you wrote is wrong before you check the arithmetic.